Final Year Project titled

"Speech To Text using CNN-LSTM"

AY 25-26

Team Details

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AGENDA

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Motivation

- Automatic Speech Recognition is fundamental for modern human-computer interaction, but maintaining accuracy in noisy environments remains a critical challenge limiting real-world usability
- Real-world applications like voice assistants, call-center automation, and smart home devices operate in noisy environments with background noise from fans, traffic, and conversations, not in sound-proof labs
- Traditional systems fail in noisy conditions, while commercial cloud-based solutions are expensive, require constant internet connectivity, and use proprietary datasets unsuitable for academic research
- There is a gap for practical, deployable solutions that can be trained and run on standard hardware without requiring cloud services or expensive infrastructure
- Our problem statement addresses this gap by designing and implementing an automatic speech-to-text system with improved robustness to environmental variations and noise, using a computationally efficient deep-learning based model suitable for CPU-only training environments

ABSTRACT

- The proposed approach utilizes Log-Mel Spectrogram features to represent speech signals, effectively capturing perceptually significant spectral characteristics. Convolutional layers extract robust local patterns from spectrograms, while bidirectional LSTM layers learn temporal relationships across speech sequences. The model is trained on a custom dataset comprising approximately 1000 voice samples, with extensive data augmentation through noise injection and SpecAugment techniques. Training uses Connectionist Temporal Classification loss, enabling end-to-end learning without explicit alignment between audio frames and text labels.
- Experimental evaluation confirms that the proposed CNN-LSTM framework delivers enhanced transcription performance with improved tolerance to noise compared to conventional baseline models. System effectiveness is assessed using Word Error Rate, Character Error Rate, and recognition accuracy across different noise levels. The results validate that the proposed method achieves a favorable balance between recognition accuracy, noise robustness, and computational efficiency, demonstrating potential for deployment in real-world speech recognition applications

INTRODUCTION

- Automatic Speech Recognition has evolved from traditional Hidden Markov Model-Gaussian Mixture Model systems to modern deep learning approaches. Traditional systems required extensive manual work including feature engineering, pronunciation dictionaries, and forced alignment, working well only in quiet environments.
- Modern deep learning approaches enable end-to-end training without manual alignment, learning optimal feature representations automatically. Our project focuses on improving speech recognition robustness to environmental variations using a hybrid CNN-LSTM architecture.
- The system processes raw audio through preprocessing, converts it to Log-Mel Spectrogram features with 80 mel frequency bins, passes through CNN layers for local pattern extraction, processes through bidirectional LSTM layers for temporal modeling, and outputs character probabilities decoded using CTC algorithm. The complete pipeline enables real-time speech recognition on CPU hardware, making it suitable for practical deployment scenarios

LITERATURE SURVEY

S.No	Paper title	Year of Publication	Methodology used	Advantage	Disadvantage
1					
2					
3					

LITERATURE SURVEY

4					
5					
6					
7					

LITERATURE SURVEY

1					
2					
3					
4					

PROPOSED METHODOLOGY

- **Dataset and Data Pipeline:** Our system uses a custom dataset comprising approximately 1000 voice samples stored in the data/raw directory, referenced through CSV manifest files in data/manifests containing audio paths and corresponding text transcripts. The dataset.py module loads these files and preprocesses audio by resampling to 16 kHz, converting stereo to mono, removing silence, and normalizing amplitude.
- **Feature Extraction:** The features.py module extracts Log-Mel Spectrogram features using 80 mel bins with window size of 400 samples and frame shift of 160 samples. These features effectively capture perceptually significant spectral characteristics of speech signals.
- **Model Architecture and Pipeline:** The features are passed through our CNN-LSTM model implemented in model.py, consisting of a CNN encoder with two convolutional layers of 64 filters each, followed by bidirectional LSTM layers with 256 hidden units, and a linear projection layer. The model outputs character probabilities decoded using CTC algorithm in decode.py. The entire system is trained using train.py, evaluated using evaluate.py, with results visualized through plots.py and deployed via web interface in web/app.py.

PROPOSED METHODOLOGY

- **Preprocessing and CNN Encoder:** Raw audio input is preprocessed by resampling to 16 kHz, converting stereo to mono, removing silence, and normalizing amplitude. The preprocessed audio is converted to Log-Mel Spectrogram features with 80 mel bins. The CNN encoder processes spectrograms through two convolutional layers with 64 filters each of size 3×3, using padding=1 to preserve temporal resolution. Each filter learns to recognize different patterns like formants for vowel sounds, energy bursts for consonants, and spectral transitions. The CNN output is reshaped from image format to sequence format for LSTM processing.
- **Bidirectional LSTM and Output Layer:** The bidirectional LSTM processes this sequence through time, learning how CNN features connect to form words. The forward LSTM reads left-to-right while the backward LSTM reads right-to-left, and at each time step both outputs are concatenated giving 512 dimensions. This allows the model to see both preceding and following context, improving recognition accuracy. The LSTM output is passed through a linear layer projecting from 512 dimensions to 29 dimensions, representing our vocabulary. Log-Softmax converts these to log-probabilities for CTC loss computation during training or CTC decoding during inference.

PROPOSED METHODOLOGY

```python
class CNNLSTMCTC(nn.Module):
    def __init__(self, vocab_size: int, n_mels: int = 80, hidden_size: int = 256):
        super().__init__()
        self.encoder = CNNSpeechEncoder(n_mels=n_mels, cnn_channels=64)
        input_size = 64 * n_mels  # 64 * 80 = 5120
        self.lstm = nn.LSTM(
            input_size=input_size,
            hidden_size=hidden_size,  # 256 per direction
            num_layers=2,
            dropout=0.3,
            bidirectional=True,
        )
        self.fc = nn.Linear(hidden_size * 2, vocab_size)  # 512 → 29
        self.log_softmax = nn.LogSoftmax(dim=-1)

    def forward(self, features: torch.Tensor) -> torch.Tensor:
        enc = self.encoder(features)  # (T, batch, 5120)
        out, _ = self.lstm(enc)  # (T, batch, 512)
        logits = self.fc(out)  # (T, batch, 29)
        log_probs = self.log_softmax(logits)
        return log_probs
```

PROPOSED METHODOLOGY

- **Model Initialization:** The CNNLSTMCTC class in model.py combines all components we discussed. The encoder processes spectrograms through two convolutional layers with 64 filters each. The LSTM has 256 hidden units per direction, 2 layers, dropout of 0.3, and bidirectional processing. The linear layer projects from 512 dimensions to 29 dimensions representing our vocabulary.

  Forward Pass: In the forward pass, features go through the CNN encoder which extracts patterns and reshapes them to sequence format. The LSTM processes this sequence bidirectionally, capturing temporal dependencies. The linear layer converts LSTM outputs to character probabilities, and Log-Softmax converts them to log-probabilities for CTC loss. This end-to-end architecture learns to map audio features directly to text

RESULTS AND DISCUSSION

- **Evaluation Metrics:** System effectiveness assessed using Word Error Rate (WER), Character Error Rate (CER), and overall recognition accuracy across different noise levels
- **Clean Speech Performance:** System achieves low WER and CER, indicating the model matches reference transcripts well
- **Moderate Noise (20 dB SNR):** Performance is very close to clean conditions, showing improved robustness compared to baseline systems
- **Higher Noise (10 dB SNR):** Some degradation occurs but our CNN-LSTM model still outperforms baseline DNN and CNN models consistently
- **Very High Noise (5 dB SNR or below):** Performance degrades further, representing an area for future improvement
- **Architecture Benefits:** CNN-LSTM maintains better performance than pure CNN or pure LSTM models because CNN layers extract robust local features less sensitive to noise, while bidirectional LSTM provides contextual information that helps correct errors
- **Data Augmentation Impact:** Extensive data augmentation including SpecAugment and speed perturbation contributes significantly to robustness by exposing the model to diverse noise conditions during training

RESULTS AND DISCUSSION

- **Baseline Comparison:** Our hybrid CNN-LSTM architecture outperforms traditional DNN-based models and pure CNN models across different noise levels
- **Noise Robustness:** Baseline models show significant performance degradation in noisy conditions, while our proposed system maintains relatively stable performance
- **Complementary Strengths:** CNNs excel at extracting local spectral-temporal patterns robust to noise, while LSTMs model long-term dependencies that help recover from noise-induced errors
- **End-to-End Training:** CTC loss allows the model to learn optimal feature representations and alignments automatically, without requiring manual feature engineering or forced alignment
- **Web Interface:** System includes a web-based interface implemented in web/app.py that enables real-time speech recognition through microphone input, demonstrating practical applicability on CPU hardware
- **Evaluation Framework:** The evaluation framework in evaluate.py computes WER and CER metrics, while plots.py generates visualization figures showing training curves, confusion matrices, and performance comparisons

CONCLUSION

- **Project Summary:** Complete deep-learning based speech-to-text system designed, implemented, and evaluated in a VTU academic setting
- **Architecture Components:** Log-Mel Spectrogram features with 80 mel bins, CNN encoder with two convolutional layers of 64 channels each, bidirectional LSTM with 2 layers and 256 hidden units, and CTC loss for end-to-end training
- **Data Augmentation:** SpecAugment and speed perturbation used to improve robustness
- **Performance:** System demonstrates improved performance in various acoustic conditions and can be trained on normal CPUs
- **Validation:** Approach validated both quantitatively through WER and CER metrics at different noise levels and qualitatively through a live web-based microphone demo
- **Results:** Hybrid CNN-LSTM architecture achieves a favorable balance between recognition accuracy, improved noise tolerance, and computational efficiency, showing promise for deployment in real-world speech recognition applications

FUTURE WORK

- **Transfer Learning:** Explore transfer learning from large pre-trained speech models and fine-tuning on task-specific datasets to reduce training time and improve robustness in noisy and low-resource conditions
- **Language Adaptation:** Adapt the system to new languages, accents, or domain-specific vocabularies
- **Speech Enhancement:** Integrate advanced speech enhancement and noise suppression modules ahead of the feature extractor to improve performance under extreme acoustic interference
- **Alternative Representations:** Experiment with alternative acoustic representations such as learned filterbanks or multi-resolution spectrograms to reveal useful trade-offs between accuracy, invariance, and computational cost
- **Model Optimization:** Optimize architecture for streaming and embedded use through model compression, parameter quantization, and structured pruning to reduce memory footprint and inference latency
- **Deployment:** Enable deployment on mobile devices and edge platforms
- **Broader Evaluation:** Conduct broader empirical evaluation on conversational speech, spontaneous dialogue, and multilingual corpora to provide a clearer picture of the system's generalization capabilities

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TIMELINE

OCTOBER

Problem Definition and Requirement Analysis

- Identified challenge of robust speech recognition in noisy environments
- Analyzed requirements for CPU-based training and deployment
- Defined project scope and objectives

NOVEMBER

Literature Survey and Design Phase

- Conducted comprehensive literature review of existing ASR approaches
- Designed hybrid CNN-LSTM architecture for speech recognition
- Planned data collection and preprocessing pipeline

DECEMBER

Implementation, Evaluation, and Documentation

- Implemented complete system using PyTorch framework
- Trained model on custom dataset of approximately 1000 voice samples
- Evaluated performance and completed documentation

Thank you